

KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2019/2020
SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR
OF SCIENCE (ENGINEERING)
ECU 107: ENGINEERING MATHEMATICS IV

DATE: FRIDAY 11TH DECEMBER 2020

TIME: 11.00 A.M. – 1.00 P.M.

INSTRUCTIONS: ANSWER QUESTION ONE AND ANY OTHER TWO

QUESTION ONE (30 MARKS)

a) Evaluate the following limits

i) $\lim_{x \rightarrow \infty} \left(\frac{1}{x} + 1 \right) \left(\frac{5x^2 - 1}{x^2} \right)$

ii) $\lim_{\theta \rightarrow 0} \frac{\theta \sin \theta}{1 - \cos \theta}$

b) Find y' given that

i) $y = \left(\frac{2x}{1-x^2} \right)^{-2/3}$

ii) $y = \sin^{-1}(\cos x)$, simplify your answer completely as possible.

c) Find equation of the tangent line to the curve whose parametric equations are $x = e^{-2t}$, $y = \sin t$ at the point $t = 0$.

d) Find the values of m and n that make the function $h(x) = \begin{cases} mx - n; & x < 1 \\ 5; & x = 1 \\ 2mx + n; & x > 1 \end{cases}$

e) Evaluate the definite integral $\int_0^1 2xe^{x^2} dx$

f) A rectangular model of area $100m^2$ is to have a minimum perimeter. Show that the model has to be a square.

g) Find the area bounded by the curve $y = 2x^2 - 3x + 2$ and the ordinates $x = 0$ and $x = 2$.



(2mks)

(3mks)

(4mks)

(3mks)

(4mks)

(4mks)

(3mks)

(4mks)

(3mks)

QUESTION TWO (20 MARKS)

a) Apply the definition of a derivative to differentiate $y = \cos x$.

b) Find $\frac{dy}{dx}$ for;

i) $y = e^{\cos x} \ln(3x)$

ii) $y = x^2 \sec^2 3x$

iii) $y = \frac{\ln(\sinh x + \cosh x)}{e^{2x}}$

(4mks)

(4mks)

(6mks)

QUESTION THREE (20 MARKS)

a) Evaluate

(5mks)

i) $\int x^2(x^3 + 5)^{1/2} dx$

(5mks)

ii) $\int_1^3 \pi(4x - x^2 - 3)^2 dx$

b) Find the point(s) at which the function $y = \frac{x+2}{x^2-4}$ is not defined. Tell whether the point is a removable point of discontinuity or not.

(5mks)

c) Find the area enclosed by the curve $y = 4x - x^2$ and the line $y = 3$.

(5mks)

QUESTION FOUR (20 MARKS)

a) Use Trapezoidal and Simpson's rules with five ordinates to approximate $\int_0^1 \frac{4}{1+x^2} dx$.

(12mks)

b) Expand $\ln x$ as a Taylor's series about the point $x = a$ up to the term containing $(x - a)^3$. Use the expansion to approximate $\ln 1.01$ correct to 5 d.p.

(9mks)

QUESTION FIVE (20 MARKS)

a) Find y'' for the curve $y + xy + y^2 + 7 = 0$

(6mks)

b) Discuss the nature of the stationary points on the curve $y = x^5 - 5x^4 + 5x^3 - 1$.

(8mks)

c) Use differentials to approximate $\sin 60^\circ 2'$

(6mks)

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (CIVIL ENGINEERING)

ECU 107: ENGINEERING MATHEMATICS IV

DATE: THURSDAY, 11TH APRIL, 2019

TIME: 11- 0AM - 1.00 PM

INSTRUCTIONS:

ANSWER QUESTION ONE AND ANY OTHER TWO.

QUESTION ONE (30 MARKS)

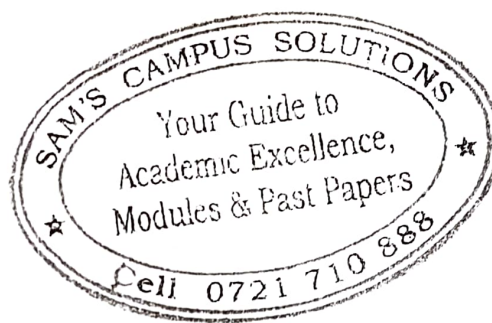
a) Evaluate the following limits

i. $\lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x}$

(4marks)

ii. $\lim_{x \rightarrow 0} \frac{\sin 2x}{x \cos 3x}$

(3marks)



b) Find the derivatives of the following function

i. $y = \ln \left(\sqrt{\frac{1-x^2}{1+x^2}} \right)$

(4marks)

ii. $y = \sin^{-1}(3x) + xe^{-x}$

(3marks)

c) Find $\frac{d^2y}{dx^2}$ given that $y = \sin 2\theta$ and $x = e^{-\theta}$

(3marks)

d) Test the continuity of the following function at $x = -1$

$$g(x) = \begin{cases} 5-2x, & \text{if } x < -1 \\ 2x, & \text{if } x \geq -1 \end{cases}$$

(4 marks).

e) Suppose that a ball is being deflated at the rate of 12cm^3 per minute. How fast is its radius reducing when the radius is 6cm?

(3marks)

f) Find the area enclosed by the curve $y = x(x-1)(x-2)$ and the x -axis

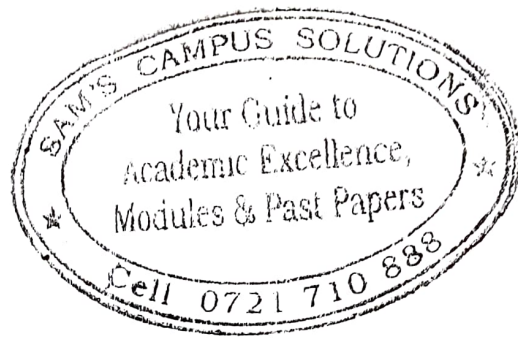
(3marks)

g) Evaluate $\int \frac{5x}{\sqrt[3]{2x^2+1}} dx$

(3marks)

QUESTION TWO (20 MARKS)

- a) Apply the definition of a derivative to find derivative of $f(x) = \frac{1}{4x-2}$ (5mks)
- b) Differentiate the following with respect to x (3mks)
- i. $y = \ln(\sinh 2x)$ (4mks)
- ii. $y = e^{-x} \tan 3x + x^2 \sec 4x$ (3mks)
- iii. $y = 10^x$
- c) Evaluate the value of $\frac{dy}{dx}$ at the point $x = -1$ given that $y^3 - x^2y - 3x - 3 = 0$ (5mks)



QUESTION THREE (20 MARKS)

- a) An open box is to be made from a 16 inch by 30 inch piece of card board by cutting out squares of equal size from the four corners and bending up the sides. What size should the squares be to obtain a box with the largest volume? (5 marks)
- b) Sketch the curve $y = x^3 - 6x^2 + 9x + 2$. (5 marks)
- c) Find the area enclosed by the curves $y = 12 - 3x^2$ and $y = 4 - x^2$ (5 marks)
- d) Use the equation $y = \sqrt{x} + \frac{1}{\sqrt{x}}$ to approximate $\sqrt{99.7}$ (5 marks)

QUESTION FOUR (20 MARKS)

- a) Find the equations of tangent and normal to the curve $y^3 + yx^2 + x^2 - 3y^2 = 0$ at $(0,3)$ (6 marks)
- b) The position of a body at a time t is $S = t^3 - 4t^2 - 3t$. Find the acceleration each time the velocity is zero. (4 marks)
- c) Differentiate the following functions
- i. $y = \frac{x^2 - 5}{\sqrt{x^3 + 7}}$ (5 marks)
- ii. $y = e^{\sin x} \tan^{-1}(x)$ (5 marks)

QUESTION FIVE (20 MARKS)

- a) Gas is escaping from a spherical balloon at a rate of $900 \text{ cm}^3 \text{ s}^{-1}$. How fast is the Surface area shrinking when the radius is 360 cm? (5 marks)

b) Evaluate the following definite integral $\int_1^2 8x\sqrt{3x^2 - 8} \, dx$

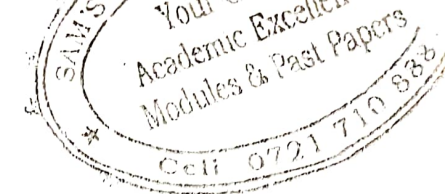
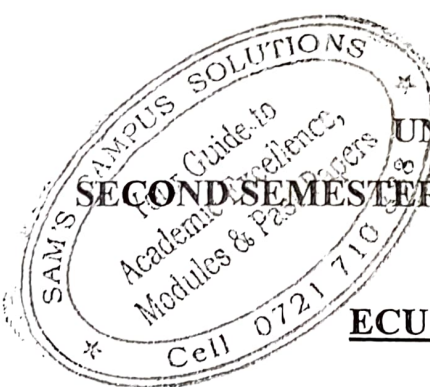
c) Find the values of constants m and n which makes the following function continuous for all values of x

$$f(x) = \begin{cases} mx - n, & \text{for } x \leq -1 \\ 2x^2 + 2mx + n, & \text{for } -1 < x \leq 1 \\ 4, & \text{for } x > 1 \end{cases}$$

(6 marks)

d) Given that $x = 2\cos t$, $y = 5\sin t$ find and $\frac{d^2 y}{dx^2}$

(3 marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

**SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE**

ECU 107: ENGINEERING MATHEMATICS IV

DATE: Friday 24th August 2018

TIME: 8.00a.m - 10.00a.m

INSTRUCTIONS:

Answer question ONE and any other TWO.

Q1. - [Compulsory - 30 marks]

(a) Investigate the continuity of the following function at the point $x = 2$.

$$f(x) = \begin{cases} \frac{3}{2}x, & \text{for } x < 2; \\ x^2 + 5, & \text{for } x \geq 2 \end{cases} \quad \text{(3 marks)}$$

(b) Evaluate the limits;

i. $\lim_{x \rightarrow 5} \frac{\frac{1}{5} - \frac{1}{x}}{x^2 - 25}$ (3 marks)

ii. $\lim_{x \rightarrow \infty} (x - \sqrt{x})$. (3 marks)

(c) Find the equation of the tangent line to the curve $x^4 + y^4x - 2x + 1 = 0$ at the point $(1, -1)$. (3 marks)

(d) Differentiate with respect to x each of the following:

i. $y = \frac{x^2 + 4}{\sqrt{x - 3}}$. (3 marks)

ii. $y = e^{-x} \sin^{-1}(x)$. (3 marks)

(e) A ball is dropped into a calm pond causing ripples in the form of concentric circles. If the radius of the outer circle is increasing at the rate 12cm per minute, how fast is the total area A of the outer circle changing when the radius is 48cm ? (3 marks)

(f) Evaluate the following definite integral: (3 marks)

$$\int_1^2 x^2 e^{x^3-1} dx.$$

(g) Find the area bounded by the graph $y = -x^2 + 3x$, the x -axis and the vertical lines $x=0$, $x = 3$. (3 marks)

- (h) The position of a body at a time t is $s = t^3 - 4t^2 - 3t$. Find the acceleration each time the velocity is zero. (3 marks)

Q2. [Optional-20 marks]

- (a) Find the area bounded by the curves $y = x^3$ and $y = 2x^2 - x^3$. (5 marks)

- (b) Evaluate the following limits:

i. $\lim_{x \rightarrow 0} \frac{\tan x}{\tan 6x - \tan 8x}$ (4 marks)

ii. $\lim_{x \rightarrow 0} (\sin x)^x$ (5 marks)

- (c) Find the values of constants A and B which makes the following function continuous for all values of x ? (6 marks)

$$f(x) = \begin{cases} Ax - B, & \text{for } x \leq -1; \\ 2x^2 + 2Ax + B, & \text{for } -1 < x \leq 1; \\ 4, & \text{for } x > 1. \end{cases}$$

Q3. [Optional-20 marks]

- (a) Differentiate $y = \frac{1}{x^2 + 1}$ from first principles.

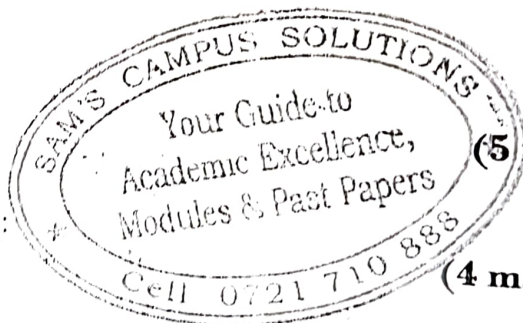
- (b) Differentiate with respect to x each of the following:

i. $y = \sinh^{-1}(\ln(x^2 + 1))$.

ii. $y = 5^{\sec x^2}$.

iii. $y = e^{x^2+y^2} + \cosh y$.

- iv. Find the points at which the graph of the of the equation $x^2y^2 - 2x = 3$ has a vertical or horizontal tangent line. (4 marks)



(5 marks)

(4 marks)

(4 marks)

(3 marks)

(4 marks)

Q4. [Optional- 20 marks]

- (a) Find the Stationary points of the curve $f(x) = 15x^4 - 10x^3 - 45x^2$. Investigate their nature and sketch the curve of $f(x)$. (11 marks)

- (b) 46200cm^2 of material is available to make a cylindrical water tank with an open top. Find the radius and the height of the tank with the largest volume.

(take $\pi = \frac{22}{7}$)

(9 marks)

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
IN BIOMEDICAL ENGINEERING

ECU 107: ENGINEERING MATHEMATICS IVDATE: Monday, 12th June, 2017

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS:Answer question **ONE** and any other **TWO**.

Q.1 (30 MARKS)

a) Evaluate the limits.

$$\text{i) } \lim_{x \rightarrow -1} \frac{x+1}{\sqrt{5+x}-2}$$

(3 marks)

$$\text{ii) } \lim_{x \rightarrow \infty} \frac{4x^2 - 2x + 3}{x^2 - 8x + 5}$$

(3 marks)

b) Differentiate with respect to x the following:-

$$\text{i) } y = \frac{\sqrt{x+2}}{x^2+4}$$

(3 marks)

$$\text{ii) } y = \frac{x}{2} (\sin(\ln x) + \cos(\ln x))$$

(3 marks)

c) Find the equation of the normal line to the curve $x^4 - x^2y + y^4 = 1$ at the point $(-1, 1)$.

(3 marks)

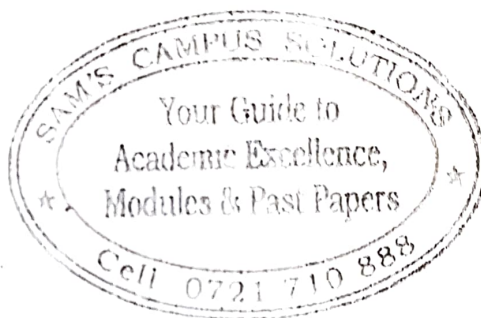
d) Find the values of a that make $f(x)$ continuous for all real x

$$f(x) = \begin{cases} e^{2x}, & x < 0 \\ x - a, & x \geq 0 \end{cases}$$

(3 marks)

c) Find $\frac{d^2y}{dx^2}$ if $x = e^{-t}$, $y = t^3$

(3 marks)



- f) A rubbish heap in the shape of a cube is being compacted. Given the volume decreases at the rate of 2 cubic inches per minute, find the rate of change of an edge. (3 marks)

g) Evaluate the definite integral $\int_1^2 x^3 e^{x^4} dx$ (3 marks)

- h) Find the area bounded by the graph $y = 2x^2 - 3x + 2$, the x -axis and the vertical lines $x = 0$, $x = 2$. (3 marks)

Q.2 a) Differentiate $y = \sqrt{x-2}$ from the first principles. (8 marks)

- b) Differentiate with respect to x in each of the following:

i) $y = \ln(\cosh(x^2 + 2))$ (4 marks)

ii) $y = x^2 \tan^{-1}(e^{3x})$ (4 marks)

iii) $y = (\cos x)^2$ (4 marks)

- Q.3 a) Find the area under the curve

$$y = \frac{x}{2 + x^2}$$

from $x = 0$ to $x = 4$ (5 marks)

- b) Find the values of a and b that make $f(x)$ continuous for all real x .

$$f(x) = \begin{cases} be^x + a + 1 & x \leq 0 \\ ax^2 + b(x + 3) & 0 \leq x \leq 1 \\ a \cos(\pi x) + 7x & x > 1 \end{cases} \quad (8 \text{ marks})$$

- c) Evaluate the following integrals.

i) $\int \frac{x^6 + 3x^2 - 2}{x^4} dx$ (4 marks)

ii) $\int \frac{\sin^3 x}{\sqrt{\cos x}} dx$ (4 marks)

- Q.4 a) Find the critical (stationary) points of the curve $f(x) = x^4 - 4x^3 + 4x^2$ and investigate their nature hence sketch the graph of $f(x)$. (10 marks)

- b) The concentration $C(t)$ (in mg/cm^2) of a drug in a patient's blood-stream after t hours is given by:

$$C(t) = \frac{t}{60(t^2 + 4t + 4)}$$

Find the maximum concentration in the first 4 hours after the patient receives the drug, and the time at which it occurs. (10 marks)

- a) The concentration of an average student during a 3 hour test at time t is given by

$$C(t) = 2t^3 - 3t^2 - 12t + 30.$$

When in the test is the concentration of the student maximal? (10 marks)

- b) Evaluate the following limits:

i) $\lim_{x \rightarrow -3} \frac{\frac{1}{x} + \frac{1}{3}}{x^3 + 27}$ (4 marks)

ii) $\lim_{x \rightarrow 0} \frac{\sin x}{\sin 6x - \sin 7x}$ (7 marks)

- c) The position of a body at time t is $s = t^3 - 4t^2 - 3t$.

Find the acceleration each time the velocity is zero. (3 marks)

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR
OF ENGINEERINGECU 107: ENGINEERING MATHEMATICS IVDATE: FRIDAY 15TH APRIL 2016

TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS

Answer Question ONE and any other TWO

Q1. - [Compulsory - 30 marks]

(a) Evaluate the limits;

i. $\lim_{x \rightarrow 49} \frac{x - 49}{6 - \sqrt{x - 13}}$. (3 marks)

ii. $\lim_{x \rightarrow 0} \frac{x - \cos x}{x}$. (3 marks)

(b) Differentiate with respect to x each of the following:

i. $y = \frac{\sqrt{x-1}}{x^2 + 4}$. (3 marks)

ii. $y = x^2 \cos^{-1}(x)$. (3 marks)

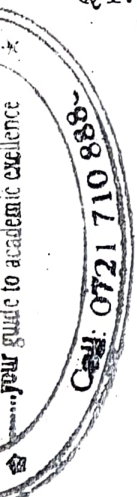
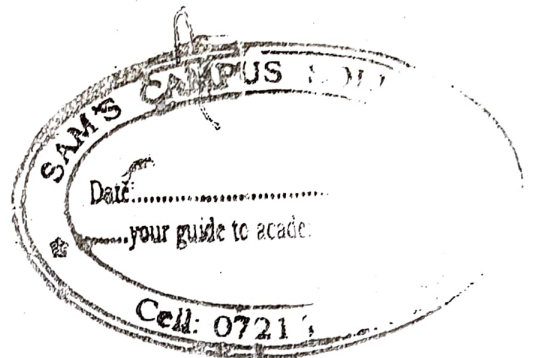
(c) Find the equation of the normal line to the curve $x^2y + y^4 - 2x - 4 = 0$ at the point $(-1, 1)$. (3 marks)(d) Find the values of the constants A and B that makes the following function continuous for all values of x . (4 marks)

$$f(x) = \begin{cases} Ax - B, & \text{for } x < 1; \\ 5; & \text{for } x = 1; \\ 2Ax + B, & \text{for } x > 1. \end{cases}$$

(e) Find $\frac{d^2y}{dx^2}$ if $x = 2t^2 + 3$ and $y = e^{-t}$. (3 marks)(f) Suppose that a ball is being inflated at the rate 12cm^3 per minute. How fast is the radius of the ball increasing when the radius is 6cm . (2 marks)

(g) Evaluate the following definite integral: (3 marks)

$$\int_1^2 x^2 e^{x^3} dx.$$



- (ii) Find the area bounded by the graph $y = 2x^2 - 3x + 2$, the x-axis and the vertical lines $x=0$, $x=2$. (3 marks)

Q2. [Optional-20 marks]

- (a) Differentiate $y = \sqrt{x-1}$ from the first principles. (8 marks)
- (b) Differentiate with respect to x each of the following:
- $y = \ln(\cosh(x^2 + 1))$. (4 marks)
 - $y = x^3 \tan^{-1}(e^{4x})$. (4 marks)
 - $y = (\sin x)^x$. (4 marks)

Q3. [Optional-20 marks]

- (a) Find the area under the curve $y = \frac{x}{\sqrt{2x+1}}$ from $x=0$ to $x=4$. (5 marks)
- (b) For what values of A and B is the following function continuous for all values of x ? (8 marks)

$$f(x) = \begin{cases} Ax - B, & \text{for } x \leq -1; \\ 2x^2 + 2Ax + B, & \text{for } -1 < x \leq 1; \\ 4, & \text{for } x > 1. \end{cases}$$

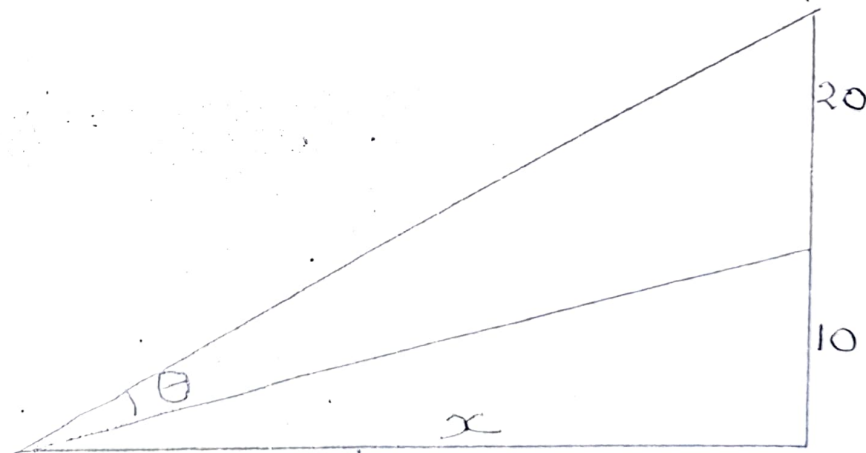
$$\begin{aligned} x = -1 & \quad -A - B = 4 \\ 2(-1)^2 + 2A(-1) + B &= 4 \\ -A - B &= 4 \\ 2(-1)^2 + 2A(-1) + B &= 4 \end{aligned}$$

- (c) Evaluate the following integrals:

- $\int \frac{x^5 + 2x^2 - 1}{x^4} dx$. (4arks)
- $\int \frac{\cos^3 x}{\sqrt{\sin x}} dx$. (4marks)

Q4. [Optional-20 marks]

- (a) A movie screen on a wall is 20ft high and 10ft above the floor. At what distance x from the front of the room should you position yourself so that the viewing angle θ of the movie screen is as largest as possible? See diagram below. (10 marks)



(b) Evaluate the following limits:

i. $\lim_{x \rightarrow -2} \frac{\frac{1}{x} + \frac{1}{2}}{x^3 + 8}$. (4 marks)

ii. $\lim_{x \rightarrow 0} (\sin x)^x$. (3 marks)

(c) The position of a body at time t is $s = t^3 - 4t^2 - 3t$. Find the acceleration each time the velocity is zero. (3 marks)

Q5. [Optional- 20 marks]

(a) Find the Stationary points of the curve $f(x) = x^4 - 2x^2 - 2$ and investigate their nature. Hence sketch the graph of $f(x)$. (10 marks)

(b) A metal box with a square top and bottom of equal size is to have a volume of 4000cm^3 . The material for the top and bottom of the box is to cost Ksh. 12 per square cm and the material for the sides is to cost Ksh. 150 per square cm. Find the dimensions of the most economical box. (10 marks)

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR
OF SCIENCEECU 107: ENGINEERING MATH IVDATE: Tuesday, 21st April 2015

TIME: 11.00 a.m. – 1.00 p.m.

INSTRUCTIONS:Answer Question 1 and any other two questions

1. Differentiate the following equations

i. From first principles $y = 4x^3 + 5x - 3$ (4 marks)

ii. $y = x^2 \cos^{-1}(x)$ (5 marks)

iii. $y = x^3 \cos 3x \ln x$ (4 marks)

iv. $\frac{3\sqrt{\theta^3}}{2 \sin 2\theta}$ (5 marks)

v. $y = 3 \tan^4 3x$ (4 marks)

vi. $(\tan x)^{\sec x}$ (4 marks)

vii. $x = a(\cos t + \log \tan(t/2)), y = a \sin t$; Find $\frac{dy}{dx}$ (4 marks)

2. i. Use the Newton's method to determine x_2 , for $f(x) = x \cos(x) - x^2$; if $x_0 = 1$ (6marks)

ii. Locate the turning point on the following curve and determine whether the point is a minimum or a maximum point $y = 4\theta + e^{-\theta}$ (6marks)

iii. Find the equation of the normal line to the curve $x^2y + y^4 - 2x - 4 = 0$ at the point $(-1, 1)$ (3 marks)

iv. Find the volume of solid generated by the region bounded by the curve $y = x\sqrt{4 - x^2}$ and the line $y = 0$ about the y axis (5 marks)

- i. The displacement x cm of the slide valve of an engine is given by;
 $x = 2.2 \cos 5\pi t + 3.6 \sin 5\pi t$. Evaluate the velocity in m/s when the time
 $t = 30ms$ (6 marks)

- ii If $f(x) = \begin{cases} -x & x < 0 \\ x & 0 < x < 1 \\ 2-x & x > 1 \end{cases}$; show that $f(x)$ is continuous at $x = 0$; and also
 at $x = 1$ (8 marks)

- iii The pressure p of the atmosphere at height h above the ground level is
 given by $p = p_0 e^{-\frac{h}{c}}$, where p_0 is the pressure at the ground level and c is a
 constant. Determine the rate of change of pressure with height when
 $p_0 = 1.013 \times 10^5$ pascals and $c = 6.05 \times 10^4$ at 1450 m. (6 marks)

4. i. Determine the area bounded by; the graph $y = 2x^2 - 3x + 2$, the x axis and the
 vertical lines $x = 0, x = 2$ (4 marks)

- ii Integrate a. $\int \left(7 - \frac{6 \sin^2 \theta}{\sin^2 \theta} \right) d\theta$ (4 marks)

b. $\int 3(8y - 1)e^{4y^2 - y} dy$ (4 marks)

c. $\int_0^{\infty} \frac{2}{1+x^2} dx$ (4 marks)

d. $\int \frac{2t^3 + 1}{(t^4 + 2t)^3} dt$ (4 marks)

5. i. Determine the volume of the solid obtained by rotating the region bounded by;
 $y = 2\sqrt{x-1}$ and $y = x-1$, about line $x = 6$ (8 marks)

- ii. An open rectangular box with square ends is fitted with an overlapping lid which
 covers the top and the front face. Determine the maximum volume of the box if 6
 m^2 of metal are used in its construction (12 marks)

UNIVERSITY EXAMINATIONS 2013/2014

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (PETROLEUM ENGINEERING, ENERGY ENGINEERING, ELECTRICAL
AND ELECTRONIC ENGINEERING, BIO-MEDICAL ENGINEERING, BY SYSTEMS,
CIVIL ENGINEERING AND MECHANICAL ENGINEERING)

ECU 107: ENGINEERING MATHEMATICS

DATE: Thursday 17th April 2014

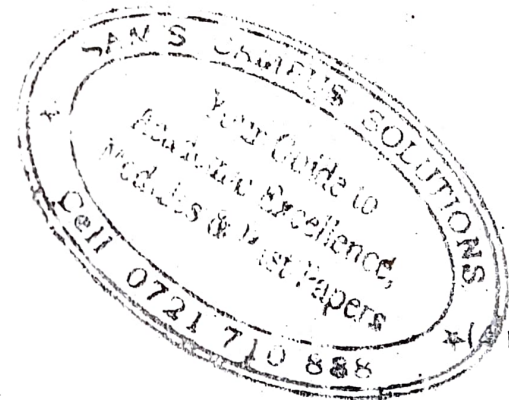
TIME: 8.00a.m -10.00a.m

INSTRUCTIONS

Question one is compulsory.

Answer Any Other Two Questions

Question one (30marks)



(4 marks)

(4marks)

(3 marks)

(3 marks)

(5 marks)

(5 marks)

(6 marks)

a) Compute the derivative of the following functions.

i) $y = e^{-3x} \sin(2x)$

ii) $f(x) = \ln\left(\frac{x+1}{x-1}\right)$

iii) $y = \sin^3 x + \cos^3 x$

✓ iv) $x^3 + y^3 + 3xy = 8$

v) $y = \sin^{-1}\left\{\frac{x}{a}\right\}$

vi) $x = \frac{t-3t}{1+t}, y = \frac{3+2t}{1+t}$

vii) $y = \left[\frac{x^2 + 1}{x^3(x-1)^2} \right]$

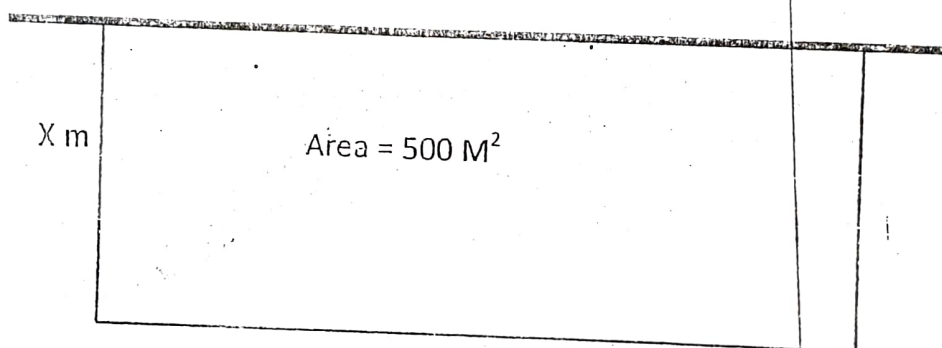
$\frac{d}{dx} (x^3 + 3x^2) = (3x^2 + 6x)$

Question two (20 marks)

- a) The parametric equation of a curve are $x = \frac{3t}{1+t}$, $y = \frac{t^2}{1+t}$. Find the equations of the tangent and normal at the point for which $t=2$. (10 marks).
- b) Find the radius of the curvature and the coordinates of the center of the curvature of the curve $y = \frac{11-4x}{3-x}$ at the point $(2, 3)$. (10 marks)

Question three (20 marks)

- a) A group of parents are planning a children's playground. In order to save on fencing, they are positioning the playground so that one side is immediately adjacent to the school building, as shown below. If the playground is to be a rectangle 500 square meters in area, what dimensions should they use if they want to use the least amount of fencing?



(8 marks)

- a) What are the dimensions of a steel tank that can hold 16 cubic meters of sludge and that uses the least material? Assume that the can is cylindrical, and is capped on both ends.

(12 marks)

Question four (20 marks)

A particle is moving along the x-axis in such a way that its position at time t for $t \geq 0$ is given by

- Show that at time $t = 0$ the particle is moving to the right. (2 marks)
- Find all values of t for which the particle is moving to the left. (2 marks)
- What is the position of the particle at time $t = 3$? (2 marks)
- When $t = 3$, what is the total distance the particle has traveled? (5 marks)
- Find the displacement of the particle on the interval $t \in [0, 3]$ (2 marks)

Q5. [Optional-20 marks]

(a) A painter finds an 26ft long ladder against a wall too heavy to lift. So she decides to slide it down by pulling it from the foot at the rate of 1.6ft/s away from the wall base. At what rate is the top of the ladder moving down the wall when the foot of the ladder is 10ft away from the wall?

(5 marks)

(b) Find $\frac{d^2y}{dx^2}$ if $x = \sin t$ and $y = e^{-t}$.

(4 marks)

(c) Evaluate the following definite integral:

$$\int_0^3 x^3 \sqrt{x^2 + 2} dx.$$

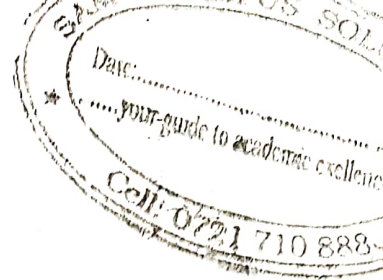
(4 marks)

(d) Find the height of a right-circular cone of maximum volume, if the sum of its height and radius of the base is 0.12m .

(7 marks)

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KENYATTA UNIVERSITY
DEPARTMENT OF MATHEMATICS



ECU 107: ENGINEERING MATHEMATICS IV –CAT I APRIL 2017

Attempt all the questions

1. Evaluate the following limits

(i) $\lim_{x \rightarrow -5} \frac{x^2 + 9x + 20}{x + 5}$

[3 marks]

(ii) $\lim_{x \rightarrow \infty} (\sqrt{x^2 - 1} - \sqrt{x^2 + 1})$

[4 marks]

(iii) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{1 - \cos 3x}$

[3 marks]



2. Use the definition of derivative to find $f'(x)$ for $f(x) = \sqrt{4x}$ [5 Marks]

3. Find the derivative of the function :

(i) $y = \ln(x^2 + \sqrt{x^4 + 1})$

[3 marks]

(ii) $y = x^2 \cos^{-1} x$

[3 marks]

4. Find the equation of the normal line to curve $y^4 - 4x^4 - 6xy = 0$ at $M(1,1)$;

[3 marks]

5. Find the values of the constant C so that the function $f(x) = \begin{cases} (x+C)^2, & x < 3 \\ 6x+C, & x \geq 3 \end{cases}$ is continuous at $x = 3$

[4 marks]

TOTAL-30 MARKS

1. Evaluate the following limits

a) $\lim_{x \rightarrow -2} \frac{\frac{1}{x} + \frac{1}{2}}{x^3 + 8}$

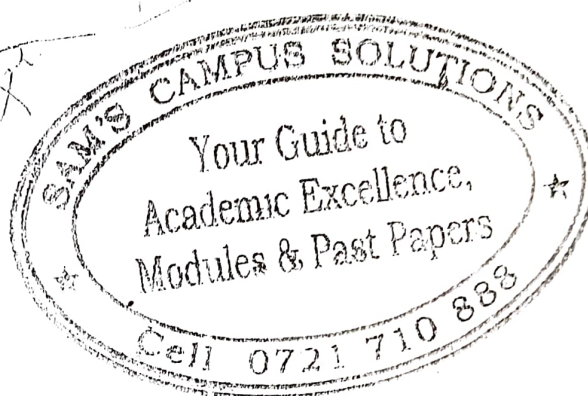
(3mks)

b) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^2}$

(3mks)

c) $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x}$

(3mks)



2. Find the values of a and b for which the following function is continuous.

$$f(x) = \begin{cases} ax - b; & x \leq -1 \\ 2x^2 + 2ax + b; & -1 < x \leq 1 \\ 4; & x > 1 \end{cases}$$

(4mks)

3. Differentiate the following functions

a) $y = \ln(\ln(x^2))$

(3mks)

b) $y = e^{2x} \tan 3x + x^2 \sec 4x$

(3mks)

c) $y = \sqrt{\frac{1-x^2}{1+x^2}}$

(3mks)

4. Find $\frac{d^2y}{dx^2}$ if $y = e^{-t}$ and $x = \sin t$

(4mks)

5. Find y'' given that $y^3 - x^2y - 3x - 3 = 7$

(4mks)

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

SUPPLEMENTARY/SPECIAL EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE

ECU 107: ENGINEERING MATHEMATICS IV

DATE: Thursday, 8th November 2018

TIME: 2.00 p.m. - 4.00 p.m.

INSTRUCTIONS:

Answer question ONE and any other TWO.



Q1. - [Compulsory - 30 marks]

(a) Evaluate the limits;

i. $\lim_{x \rightarrow 36} \frac{5 - \sqrt{x - 11}}{x - 36}$. (3 marks)

ii. $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^2}$. (3 marks)

(b) Differentiate with respect to x each of the following:

i. $y = \frac{x^2 + 4}{\sqrt{x - 1}}$. (3 marks)

ii. $y = x^2 \sin^{-1}(x)$. (3 marks)

(c) Find the equation of the normal line to the curve $y^3 - x^2y - 3x - 3 = 0$ at the point $(-1, 1)$. (3 marks)

(d) Find the values of the constants A and B that makes the following function continuous for all values of x . (4 marks)

$$f(x) = \begin{cases} Ax - B, & \text{for } x < 1; \\ -3, & \text{for } x = 1; \\ 2Ax + B, & \text{for } x > 1 \end{cases}$$

(e) Find $\frac{d^2y}{dx^2}$ if $x = 2t^2 + 3$ and $y = e^{-t}$. (3 marks)

(f) Suppose that a ball is being inflated at the rate 12cm^3 per minute. How fast is the radius of the ball increasing when the radius is 6cm . (2 marks)

(g) Evaluate the following definite integral: (3 marks)

$$\int_1^2 x^2 e^{x^3} dx.$$

- (h) Find the area bounded by the graph $y = 2x^2 - 3x + 2$, the x-axis and the vertical lines $x=0$, $x= 2$. (3 marks)

Q2. [Optional-20 marks]

- (a) Find the area under the curve $y = \frac{x}{\sqrt{2x+1}}$ from $x = 0$ to $x = 4$. (5 marks)
- (b) For what values of A and B is the following function continuous for all values of x ? (8 marks)

$$f(x) = \begin{cases} Ax - B, & \text{for } x \leq -1; \\ 2x^2 + 2Ax + B, & \text{for } -1 < x \leq 1; \\ 4, & \text{for } x > 1. \end{cases}$$

- (c) Evaluate the following integrals:

i. $\int \frac{x^5 + 2x^2 - 1}{x^4} dx$. (4arks)

ii. $\int \frac{\cos^3 x}{\sqrt{\sin x}} dx$. (4marks)

Q3. [Optional-20 marks]

- (a) Differentiate $y = \sqrt{x-1}$ from the first principles. (8 marks)

- (b) Differentiate with respect to x each of the following:

i. $y = \ln(\sinh(x^2 + 1))$. (4 marks)

ii. $y = x^3 \tan^{-1}(e^{4x})$. (4 marks)

iii. $y = (\cos x)^x$. (4 marks)

Q4. [Optional- 20 marks]

- (a) Find the Stationary points of the curve $f(x) = x^4 - 2x^2 - 2$ and investigate their nature. Hence sketch the graph of $f(x)$. (10 marks)

- (b) A metal box with a square top and bottom of equal size is to have a volume of 4000cm^3 . The material for the top and bottom of the box is to cost Ksh. 12 per square cm and the material for the sides is to cost Ksh. 150 per square cm. Find the dimensions of the most economical box. (10 marks)

Q5. [Optional-20 marks]

- (a) A movie screen on a wall is 20ft high and 10ft above the floor. At what distance x from the front of the room should you position yourself so that the viewing angle θ of the movie screen is as large as possible? See diagram below. (10 marks)

CAT 2

ATTEMPT ALL QUESTIONS

1. Given that $x = 2 \cos t$, $y = 5 \sin t$ find $\frac{dy}{dx}$ and $\frac{d^2 y}{dx^2}$ (4 marks)
2. A particle moves on the hyperbola $3x^2 - y^2 = 12$ in such a way that its y coordinate increases at a constant rate of 9 units per second. How ~~fast~~ first is its x coordinate changing when $x = 3$. (4 marks)
3. Evaluate the following integrals
 - i. $\int \frac{2x}{\sqrt[3]{2x^2 + 1}} dx$ (4 marks)
 - ii. $\int_0^{\pi} \cos x e^{\sin x} dx$ (5 marks)
4. Sketch the graph of $y = x^3 - 3x + 2$ (5 marks)
5. Use the equation $y = \sqrt{x} + \frac{1}{\sqrt{x}}$ to approximate $\sqrt{99.7}$ (4 marks)
6. The position of a body at a time t is $S = t^3 - 4t^2 - 3t$. Find the acceleration each time the velocity is zero. (4 marks)

CAT 2

ANSWER ALL QUESTIONS

$\delta x = \frac{dy}{dx}$

$\delta y = \frac{dy}{dx} \delta x$

$= y + \delta y$

1. Use the equation $y = \sqrt{x}$ to approximate $\sqrt{99.7}$. (5 marks)

2. Distinguish the stationary points of the curve $y = x^3 - 3x^2 - 9x + 25$. (6 marks)

3. A spherical balloon is to be ~~deflated~~ ^{decrease in volume} so that its radius decreases at constant rate of 15cm/min. At what rate must air be removed when the radius is 9cm? (5 marks)

4. Find the equations of tangent and normal to the curve $x^3 + y^3 = 3xy$ at the point $(\frac{3}{2}, \frac{3}{2})$. (5 marks)

$(\frac{3}{2}, \frac{3}{2})$

5. Evaluate the following indefinite integrals

i. $\int x^2 \sqrt[3]{x^3 + 5} dx$

ii. $\int \frac{x^3 - 2x^2 + 4x + 9}{3x^2} dx$



(5 marks)

(4 marks)

Q5. [Optional-20 marks]

- (a) A painter finds an 26ft long ladder against a wall too heavy to lift. So she decides to slide it down by pulling it from the foot at the rate of 1.6ft/s away from the wall base. At what rate is the top of the ladder moving down the wall when the foot of the ladder is 10ft away from the wall?

(5 marks)

- (b) Find $\frac{d^2y}{dx^2}$ if $x = \sin t$ and $y = e^{-t}$.

(4 marks)

- (c) Evaluate the following definite integral:

$$\int_0^3 x^3 \sqrt{x^2 + 2} dx.$$

(4 marks)

- (d) Find the height of a right-circular cone of maximum volume, if the sum of its height and radius of the base is 0.12m .

(7 marks)

..... E N D

(b) Evaluate the following limits:

i. $\lim_{x \rightarrow -2} \frac{\frac{1}{x} + \frac{1}{2}}{x^3 + 8}$. (4 marks)

ii. $\lim_{x \rightarrow 0} (\sin x)^x$. (3 marks)

(c) The position of a body at time t is $s = t^3 - 4t^2 - 3t$. Find the acceleration each time the velocity is zero. (3 marks)